

Abhishek Singh

B.Tech Electronics and Communication Engineering, IIT Patna (Batch of 2027)

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Education

Indian Institute of Technology Patna

2023–2027

B.Tech in Electronics and Communication Engineering (Electrical Department)

CPI: 7.0/10.0

Relevant Coursework: DSA, AI & ML, Probability & Statistics, DSP, Signals & Systems, Linear Algebra

Scholastic Achievements

- AIR 1046 in GATE (ECE), Score 662.
- AIR 5701 in JEE Advanced and AIR 4311 in JEE Main.
- Solved 1000+ DSA problems across competitive programming platforms.
- Completed extensive ML/DL coursework including Transformers, Computer Vision and AutoEncoders.

Research & Leadership Experience

Video Surveillance Research Intern, IIT Patna

May 2026– July 2026

Under Prof. Mahesh Kumar Kolekar

- Developed AutoEncoder-based anomaly detection systems for intelligent surveillance using PyTorch.
- Designed Hybrid Patch Feature Scoring for anomaly localization and scene-level anomaly detection.
- Trained and evaluated models on UCSD Pedestrian, CUHK Avenue and ShanghaiTech datasets.

Independent Research — Silicon Photonic Microring Resonators

May 2026 – June 2026

- Simulated lateral, vertical/3D, and parallel-cascaded coupling configurations using the open-source MEEP FDTD framework.
- Performed coupling-gap and critical-coupling sweeps, quality-factor (Q) analysis, and numerical convergence verification.
- Authored a citable technical report (Zenodo).

Research under Prof. Sumanta Gupta, IIT Patna

May 2026–Present

Photonics / Optical Communication Group

- Exploring photonic-electronic implementations of neural-network primitives, focusing on optical-domain multiply-accumulate (MAC) operations for silicon photonic hardware.
- Studying microring-resonator weighting and balanced-detection schemes for signed-weight optical computation.
- Working toward a photoelectronic multiply-accumulate neuron (PEMAN) architecture that combines optical-domain multiplication with electronic accumulation and activation.

Software Head, Inditronix AI Labs

April 2025 – May 2026

- Led software architecture and development for healthcare AI solutions.
- Developed and deployed production infrastructure on AWS EC2.
- Built ESP32-based IoT applications and coordinated multidisciplinary software teams.

Projects

AI Surveillance System

GitHub

- Developed AutoEncoder-based video anomaly detection framework using PyTorch.
- Designed Hybrid Patch Feature Scoring and evaluated on UCSD, CUHK Avenue and ShanghaiTech datasets.
- Built end-to-end training, inference and evaluation pipelines.

USAS Ingestor Engine

Private Repository

- Built scalable ingestion pipeline for multi-camera CCTV and DOT camera streams.
- Developed synchronization, preprocessing and storage modules for surveillance analytics.

Custom 3D Rendering Engine

GitHub

- Developed a custom real-time 3D rendering engine with gesture-based interaction using Three.js (WebGL).
- Built a spline editor and procedural scene generation framework; created and prepared 3D assets in Blender.

3D Art, Visualization & Digital Media (Blender)

Portfolio

- Produced photorealistic product renders and stylized 3D scenes in Blender, covering modeling, materials/shading, lighting and camera/depth-of-field composition.
- Created environment models, low-poly scenes and shader-driven visuals; published animated renders as short-form motion pieces.
- Skilled in video editing and digital illustration/drawing, maintaining a focused creative-technical portfolio.

Competitive Programming

Codeforces ([journey_to_grandmaster](#)): 323+ solved, Rating 1241 (Pupil), Best Rank 2295 (Div-2)

CodeChef ([abhishek_5701](#)): 174+ solved, Rating 1643 (3-Star), Best Ranks 150 (Div-3), 459 (Div-2)

LeetCode: 121+ solved

GeeksforGeeks: 183+ solved

1000+ DSA problems solved across online judges.

Kaggle

Participated in ML competitions involving healthcare, finance and environmental prediction, including Problematic Internet Use Prediction, Loan Payback Prediction and Rainfall Dataset Prediction.

Technical Skills

Languages: Python, C++, C, SQL
ML/AI: TensorFlow, PyTorch, Deep Learning, Computer Vision, CNNs, Transformers, AutoEncoders, Anomaly Detection
Tools: Linux, Git, AWS EC2, OpenCV, NumPy, Pandas, MEEP, ESP32, Three.js, WebGL, Blender

Affiliations

IEEE Student Member — IEEE Photonics Society (since June 2026).